0 Self-Grading Declaration

Based on the guidelines provided, **I self-assign a grade of 2/2 for this homework**. I have made a legitimate effort to complete 100% of the problems.

1 Assignment 1

Assignment 1

Consider a slender beam of length L, depth h (and thickness b). The beam is clamped at its base and bent around a rigid mandrel of diameter $d = 2L/\pi$, so that it wraps exactly half-way around the mandrel and that the end-sections are exactly vertical.

(a) Write down an expression for the motion that is compatible with the given drawing.

(b) Where is the neutral axis for this motion (the axis of zero normal strain during bending)?

(c) Compute the Green-Lagrange strain component E_{11} and the small-strain component ϵ_{11} for the material point $\boldsymbol{X} = (L, L/\pi)$ – the lower extreme corner of the undeformed beam. Compare and comment on your result.

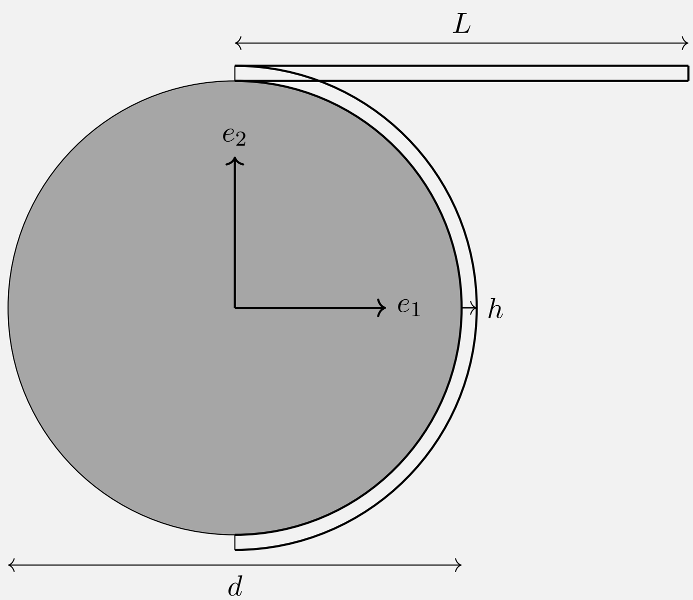

Figure 1: Reference diagram for Assignment 1.

Treat as a two-dimensional problem; see also A&G 3.13. Consider using something like Mathematica to handle the algebra.

The initial position vector $\boldsymbol{X}$, which defines the location of material points in the reference configuration based on **Fig. 1**, is given by:

$$\boldsymbol{X} = (X_1, X_2) = \left(\lambda_x,\, \frac{L}{\pi} + \lambda_y\right) \quad \lambda_x \epsilon (0; L)\, \lambda_y \epsilon (0; h)$$

The corresponding position vector $\boldsymbol{x}$ in the deformed configuration can be expressed as:

$$\boldsymbol{x} = \left(X_2 sin(\frac{\pi}{L}X_1),\, X_2 cos(\frac{\pi}{L}X_1)\right)$$

1.1 Green-Cauchy tensor

The deformation gradient $\boldsymbol{F}$ is obtained by differentiating $\boldsymbol{x}$ with respect to $\boldsymbol{X}$ [1]:

$$\boldsymbol{F} = \begin{bmatrix} X_2\frac{\pi}{L}cos(\frac{\pi}{L}X_1) & sin(\frac{\pi}{L}X_1) \\ -X_2\frac{\pi}{L}sin(\frac{\pi}{L}X_1) & cos(\frac{\pi}{L}X_1) \end{bmatrix} \quad \boldsymbol{F}^{\mathsf{T}} = \begin{bmatrix} X_2\frac{\pi}{L}cos(\frac{\pi}{L}X_1) & -X_2\frac{\pi}{L}sin(\frac{\pi}{L}X_1) \\ sin(\frac{\pi}{L}X_1) & cos(\frac{\pi}{L}X_1) \end{bmatrix}$$

The right Cauchy–Green tensor is defined as $\boldsymbol{C} = \boldsymbol{F}^{\mathsf{T}}\boldsymbol{F}$.

$$\boldsymbol{C} = \boldsymbol{F}^T\boldsymbol{F} = \begin{bmatrix} (X_2\frac{\pi}{L})^2 & 0^1 \\ 0^1 & 1^2 \end{bmatrix} \tag{1}$$

1.2 Neutral axis

The neutral axis of the beam is the one-dimensional geometric line spanning along its length (i.e. the $\boldsymbol{e_1}$ direction for the reference body). It consists of all material points where the normal strain ϵ is zero, or equivalently, where the stretch λ given by **Eq. (2)**[1] equal to 1 $\dot{}$Thus:

$$\lambda = \sqrt{\boldsymbol{n}C \cdot \boldsymbol{n}} = 1 \tag{2}$$

Replacing $\boldsymbol{n}$ by the direction $\boldsymbol{e_1}$:

$$\lambda = \sqrt{\boldsymbol{e_1}C \cdot \boldsymbol{e_1}} = 1$$

Which is an equivalent of the component C_{11} of tensor $\boldsymbol{C}$, thus:

$$X_2\frac{\pi}{L} = \pm 1 \xrightarrow{X_2>0} X_2 = \frac{L}{\pi}\,\forall X_1 \tag{3}$$

The geometric set described by **Eq. (3)** corresponds to all material points in the undeformed configuration with a coordinate $X_2 = \frac{L}{\pi} = \frac{d}{2}$. These points define a layer of fibers initially located at the *bottom surface of the beam* along its entire length. When the beam is bent around the mandrel, these fibers deform into a circular arc of radius $R = \frac{L}{\pi}$ and therefore *undergo no axial stretch*. This defines the *neutral axis* of the motion, as illustrated in **Fig. 2**.

[1] $C_{21} = C_{12} = X_2\frac{\pi}{L}cos(\frac{\pi}{L}x_1)sin(\frac{\pi}{L}x_1) - X_2\frac{\pi}{L}cos(\frac{\pi}{L}x_1)sin(\frac{\pi}{L}x_1) = 0$
[2] $sin(\frac{\pi}{L}X_1)^2 + cos(\frac{\pi}{L}X_1)^2 = 1$

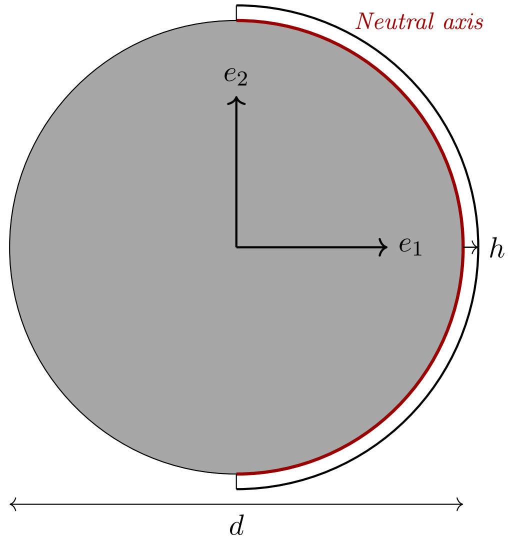

Figure 2: Neutral axis in the deformed configuration.

1.3 Green-Lagrange component

The Green-Lagrange strain tensor $\boldsymbol{E}$ is defined from $\boldsymbol{C}$. Starting from **??**:

$$\boldsymbol{E} = \frac{1}{2}(\boldsymbol{C} - \mathbf{1}) = \frac{1}{2}\begin{bmatrix} (X_2 \frac{\pi}{L})^2 - 1 & -1 \\ -1 & 0 \end{bmatrix}$$

Taking the E_{11} component in $X_1 = L$ and $X_2 = L/\pi$:

$$E_{11} = (\frac{\pi}{L}\frac{L}{\pi})^2 - 1 = 0 \tag{4}$$

1.4 Small Strain Tensor

On the other hand, the small strain tensor $\boldsymbol{\epsilon}$ is derived from the displacement gradient $\mathbf{H}$ via:

$$\boldsymbol{\epsilon} = \frac{1}{2}\left(\mathbf{H} + \mathbf{H}^\top\right), \quad \text{where} \quad \mathbf{H} = \mathbf{F} - \mathbf{I}.$$

Given the deformation gradient:

$$\mathbf{F} = \begin{bmatrix} \dfrac{\pi X_2}{L}\cos\left(\dfrac{\pi X_1}{L}\right) & \sin\left(\frac{\pi X_1}{L}\right) \\ -\dfrac{\pi X_2}{L}\sin\left(\dfrac{\pi X_1}{L}\right) & \cos\left(\frac{\pi X_1}{L}\right) \end{bmatrix},$$

so the displacement gradient is:

$$\mathbf{H} = \mathbf{F} - \mathbf{I} = \begin{bmatrix} \dfrac{\pi X_2}{L}\cos\left(\dfrac{\pi X_1}{L}\right) - 1 & \sin\left(\frac{\pi X_1}{L}\right) \\ -\dfrac{\pi X_2}{L}\sin\left(\dfrac{\pi X_1}{L}\right) & \cos\left(\frac{\pi X_1}{L}\right) - 1 \end{bmatrix}.$$

Now taking the symmetric part:

$$\boldsymbol{\epsilon} = \frac{1}{2}\left(\mathbf{H} + \mathbf{H}^\top\right) = \frac{1}{2}\begin{bmatrix} 2\left(\frac{\pi X_2}{L}\cos\left(\frac{\pi X_1}{L}\right) - 1\right) & \sin\left(\frac{\pi X_1}{L}\right) - \frac{\pi X_2}{L}\sin\left(\frac{\pi X_1}{L}\right) \\ \sin\left(\frac{\pi X_1}{L}\right) - \frac{\pi X_2}{L}\sin\left(\frac{\pi X_1}{L}\right) & 2\left(\cos\left(\frac{\pi X_1}{L}\right) - 1\right) \end{bmatrix}.$$

Simplifying:

$$\boldsymbol{\epsilon} = \begin{bmatrix} \dfrac{\pi X_2}{L}\cos\left(\dfrac{\pi X_1}{L}\right) - 1 & \dfrac{1}{2}\left(1 - \dfrac{\pi X_2}{L}\right)\sin\left(\dfrac{\pi X_1}{L}\right) \\ \dfrac{1}{2}\left(1 - \dfrac{\pi X_2}{L}\right)\sin\left(\dfrac{\pi X_1}{L}\right) & \cos\left(\dfrac{\pi X_1}{L}\right) - 1 \end{bmatrix}.$$

Evaluating the small strain tensor at $X_1 = L$ and $X_2 = \dfrac{L}{\pi}$ and substituting into the expression:

$$\boldsymbol{\epsilon} = \begin{bmatrix} 1 \cdot \cos(\pi) - 1 & \frac{1}{2}(1-1)\sin(\pi) \\ \frac{1}{2}(1-1)\sin(\pi) & \cos(\pi) - 1 \end{bmatrix} = \begin{bmatrix} -1-1 & 0 \\ 0 & -1-1 \end{bmatrix} = \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix}.$$

1.5 Conclusion

The small-strain approximation fails at this point because the deformation involves a finite (and considerable) rotation. Although the material experiences a stretch for $X_2 > d = L/\pi$, the small-strain tensor interprets this rotation as large compression. In contrast, the Green–Lagrange strain correctly accounts for the geometry and returns zero for a specific fiber that lies in the neutral axis, as demonstrated in Section 1.2.

2 Assignment 2

Assignment 2

Displacive/martensitic phase transformations[3] in metals are often characterized by the strain induced in the material when the material transforms from one phase to the other. Experimentally one can measure the atomic positions before and after the phase transformation and use this to determine the transformation strain in the material.
Consider the situation in a cubic to monoclinic phase transformation where the cubic lattice cell is defined by the three vectors

$$\boldsymbol{a}_1 = a_0 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \boldsymbol{a}_2 = a_0 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \boldsymbol{a}_3 = a_0 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix},$$

where $a_0 = 6.0\,\text{Å}$, and the monoclinic lattice cell is defined by

$$\boldsymbol{m}_1 = a \begin{pmatrix} \cos(\pi/2-\beta) - \sin(\pi/2-\beta) \\ \cos(\pi/2-\beta) + \sin(\pi/2-\beta) \\ 0 \end{pmatrix}, \quad \boldsymbol{m}_2 = c \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \boldsymbol{m}_3 = b \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix},$$

where $a = 4.55\,\text{Å},\ b = 5.45\,\text{Å},\ c = 6.02\,\text{Å}$, and $\beta = 87^\circ \times \dfrac{\pi}{180}$. All vector components have been given in the same frame (that of the cubic lattice).
Define the strain as the Biot/Bain strain $\boldsymbol{E}_{\text{Biot}} = \boldsymbol{U} - \boldsymbol{1}$ and determine the principal strains of the phase transformation; see A&G 3.4 and 3.9.
Assume that $\boldsymbol{a}_k$ maps to $\boldsymbol{m}_k$.

2.1 Gradient Tensor $\boldsymbol{F}$ Reconstruction

The description of the problems provides three different mappings of the form $\boldsymbol{F}\boldsymbol{a} = \boldsymbol{m}$ which are used to reconstruct the tensor $\boldsymbol{F}$ under the approximation that:

$$F dX_i = x_i \approx F\Delta X = \Delta x \tag{5}$$

Eq. (5) holds true when $\Delta X, \Delta x \to 0$ which is considered appropiate to the cell-scale of the problem at a material-level description. $\boldsymbol{F}$ is constructed by combining the the vectors $\boldsymbol{a_i}$ into columns of the matrix $\boldsymbol{A}$ and vectors $\boldsymbol{m_i}$ into columns of the matrix $\boldsymbol{M}$, and thus solving the linear system $\boldsymbol{F}\boldsymbol{A} = \boldsymbol{M}$ by computing $\boldsymbol{A}^{-1}$ and solving for $\boldsymbol{F}$:

$$\boldsymbol{A} = \begin{bmatrix} | & | & | \\ a_1 & a_2 & a_3 \\ | & | & | \end{bmatrix} = \begin{bmatrix} 6.0 & 0 & 0 \\ 0 & 6.0 & 0 \\ 0 & 0 & 6.0 \end{bmatrix},$$

$$\boldsymbol{M} = \begin{bmatrix} | & | & | \\ m_1 & m_2 & m_3 \\ | & | & | \end{bmatrix} = \begin{bmatrix} 4.306 & 0 & 0 \\ 4.782 & 6.020 & 0 \\ 0 & 0 & 5.450 \end{bmatrix}.$$

This leads to the linear system

$$\boldsymbol{F} = \boldsymbol{M}\boldsymbol{A}^{-1}.$$

Explicitly, with:

$$\boldsymbol{A}^{-1} = \frac{1}{6.0}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

$\boldsymbol{F}$ can be computed as:

$$\boldsymbol{F} = \begin{bmatrix} 0.7176 & 0 & 0 \\ 0.7970 & 1.0033 & 0 \\ 0 & 0 & 0.9083 \end{bmatrix}$$

Here, $\boldsymbol{F}$ denotes the deformation matrix (gradient) that maps the cubic crystal system to its monoclinic counterpart. To streamline the process, the matrix was obtained computationally using Python. For details on the implementation, refer to Section 6.

2.2 Spectral Decomposition of $\boldsymbol{C}$

The right Cauchy-Green tensor $\boldsymbol{C}$ is proven to be positive definite and symetric and thus admits spectral decomposition, which can be used to obtain the Right Strech Tensor $\boldsymbol{u}$ by polar decomposition.

By construction, $\boldsymbol{C}$ is symmetric and positive definite. These two properties ensure that $\boldsymbol{C}$ admits a *spectral decomposition*, namely

$$\boldsymbol{C} = \sum_{i=1}^{3} \lambda_i^2 \, \boldsymbol{r}_i \otimes \boldsymbol{r}_i = \boldsymbol{F}^{\mathsf{T}}\boldsymbol{F} = \boldsymbol{U}^2,$$

where λ_i^2 are the eigenvalues of $\boldsymbol{C}$ and $\boldsymbol{r}_i$ the corresponding orthonormal eigenvectors. The right stretch tensor $\boldsymbol{U}$ is obtained as the unique positive definite square root of $\boldsymbol{C}$. Using the spectral decomposition, $\boldsymbol{U}$ can be written explicitly as:

$$\boldsymbol{U} = \sum_{i=1}^{3} \lambda_i \, \boldsymbol{r}_i \otimes \boldsymbol{r}_i,$$

Note that $\boldsymbol{r}_i$ are the same eigenvectors of $\boldsymbol{C}$ but with coefficients given by the square roots of the eigenvalues[1].

In practice, the computation was carried out numerically using Python in section Section 6. The procedure consisted of:

1. Constructing the deformation gradient $\boldsymbol{F}$ by mapping cubic lattice vectors into monoclinic lattice vectors.
2. Forming the right Cauchy–Green tensor $\boldsymbol{C} = \boldsymbol{F}^{\mathsf{T}}\boldsymbol{F}$.
3. Solving the eigenvalue problem for $\boldsymbol{C}$ to obtain eigenvalues and eigenvectors.

[1] This follows because if $\boldsymbol{U}\boldsymbol{r}_i = \lambda_i \boldsymbol{r}_i$, then $\boldsymbol{C}\boldsymbol{r}_i = (\boldsymbol{U}^2)\boldsymbol{r}_i = \boldsymbol{U}(\lambda_i \boldsymbol{r}_i) = \lambda_i(\boldsymbol{U}\boldsymbol{r}_i) = \lambda_i^2 \boldsymbol{r}_i$. Hence the eigenvectors of $\boldsymbol{C}$ and $\boldsymbol{U}$ coincide, but the eigenvalues of $\boldsymbol{C}$ are the squares of those of $\boldsymbol{U}$, i.e. λ_i^2.

4. Assembling $\boldsymbol{U}$ using the formula above, i.e. combining eigenvectors $\boldsymbol{r}_i$ with the square roots of the eigenvalues.

The numerical results obtained from the spectral decomposition are summarized below. The eigenvalues of $\boldsymbol{C}$ are:

$$\lambda_i^2 = \begin{bmatrix} 1.8813 & 0.2756 & 0.8251 \end{bmatrix},$$

With corresponding eigenvectors:

$$\boldsymbol{r}_1 = \begin{bmatrix} 0.7380 \\ 0.6748 \\ 0 \end{bmatrix}, \quad \boldsymbol{r}_2 = \begin{bmatrix} -0.6748 \\ 0.7380 \\ 0 \end{bmatrix}, \quad \boldsymbol{r}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Using these, the right stretch tensor reconstructed from the spectral decomposition is given by:

$$\boldsymbol{U} = \sum_{i=1}^{3} \lambda_i \, \boldsymbol{r}_i \otimes \boldsymbol{r} = \begin{bmatrix} 0.9861 & 0.4216 & 0 \\ 0.4216 & 0.9104 & 0 \\ 0 & 0 & 0.9083 \end{bmatrix}.$$

As expected, it was verified numerically that

$$\boldsymbol{U}^2 = \boldsymbol{C}$$

within machine precision, thus confirming the theoretical result.

2.3 Conclusion

Finally, the $\boldsymbol{E}_{\text{Biot}}$ tensor is constructed:

$$\boldsymbol{E}_{\text{Biot}} = \boldsymbol{U} - \boldsymbol{I} = \begin{bmatrix} -0.0139 & 0.4216 & 0 \\ 0.4216 & -0.0896 & 0 \\ 0 & 0 & -0.0917 \end{bmatrix}.$$

Consequently, the numerical values of the principal strains obtained from taking the square root of the eigenvalues of $\boldsymbol{C}$ and subtracting 1:

$$\varepsilon_1 = 0.3716, \quad \varepsilon_2 = -0.475, \quad \varepsilon_3 = -0.09.$$

It is important to note that the principal strains correspond to the eigenvalues of the Biot strain tensor.[2]

[2] The key point is that the identity tensor can also be expressed in the eigenbasis of $\boldsymbol{U}$ as $\boldsymbol{I} = \sum_{i=1}^{3} \boldsymbol{r}_i \otimes \boldsymbol{r}_i$, since $\{\boldsymbol{r}_i\}$ form an orthonormal basis. Therefore,

$$\boldsymbol{E}_{\text{Biot}} = \boldsymbol{U} - \boldsymbol{I} = \sum_{i=1}^{3} \lambda_i \, \boldsymbol{r}_i \otimes \boldsymbol{r}_i - \sum_{i=1}^{3} 1 \cdot \boldsymbol{r}_i \otimes \boldsymbol{r}_i = \sum_{i=1}^{3} (\lambda_i - 1) \, \boldsymbol{r}_i \otimes \boldsymbol{r}_i.$$

This shows that the eigenvectors of $\boldsymbol{E}_{\text{Biot}}$ coincide with those of $\boldsymbol{U}$, while its eigenvalues are $\lambda_i - 1$, i.e. the principal strains.

3 Assignment 3

Assignment 3

Consider the small strain tensor (at a point):

$$\boldsymbol{\varepsilon} = \begin{bmatrix} 1.0 & 2.0 & 0.0 \\ 2.0 & 0.5 & 0.0 \\ 0.0 & 0.0 & -1.5 \end{bmatrix} \quad \text{mstrain.}$$

(a) What is the volumetric strain at this point?

(b) What are the maximum and minimum principal strains at this point?

(c) What is the normal strain in the direction $\mathbf{n} = (1.0,\ 1.0,\ 1.0)/\sqrt{3}$?

(d) What is the tensorial shear strain associated with the directions $\mathbf{n} = (1.0,\ 1.0,\ 1.0)/\sqrt{3}$ and $\mathbf{m} = (0.0,\ 1.0,\ -1.0)/\sqrt{2}$?

See A&G 3.11, 3.14, and 3.15[2].

(a) Volumetric strain. For infinitesimal deformations, the local volume change per unit original volume is given by the trace of the small strain tensor [1]:

$$\frac{dv - dv_R}{dv_r} = tr\boldsymbol{\epsilon} = \epsilon_{kk} = 0..$$

(b) Principal strains. Principal strains are obtained as the eigenvalues of $\boldsymbol{\epsilon}$:

$$[\boldsymbol{\varepsilon}] = [\,2.7656,\ -1.2656,\ -1.5000\,]$$

The corresponding eigenvectors (principal directions) are given by the columns of the eigenvector matrix returned by `numpy.linalg.eig`[3]. For this example, the maximum principal strain at the point analyzed is 2.76 and the minimum -1.50.

(c) Normal strain in a given direction. For a unit vector $\boldsymbol{n}$, the normal strain is:

$$\varepsilon_n = \boldsymbol{n}^{\mathsf{T}} \boldsymbol{\epsilon}\, \boldsymbol{n}.$$

For $\boldsymbol{n} = (1,1,1)/\sqrt{3}$, the result is

$$\varepsilon_{nn} = 4/3$$

(d) Tensorial shear strain. The tensorial shear strain in directions $\boldsymbol{n}$ and $\boldsymbol{m}$ is

$$\epsilon_{nm} = \boldsymbol{n}^{\mathsf{T}} \boldsymbol{\epsilon}\, \boldsymbol{m}.$$

With $\boldsymbol{n} = (1,1,1)/\sqrt{3}$ and $\boldsymbol{m} = (0,1,-1)/\sqrt{2}$, the result is

$$\epsilon_{nm} = \epsilon_{mn} \approx 1.63$$

Each step corresponds directly to one part of the assignment and was verified numerically with Python. This modular approach ensures the theoretical relations are illustrated with explicit numerical results.

4 Assignment 4

Assignment 4

Consider the displacement field of a screw dislocation aligned with the $\mathbf{e}_3$ axis:

$$u_3 = \frac{b\,\theta}{2\pi} \;\equiv\; \frac{b}{2\pi}\tan^{-1}\left(\frac{X_2}{X_1}\right),$$

where $0 \leq \theta < 2\pi$ and b is the Burgers distance.

(a) Determine the small strain field $\boldsymbol{\epsilon}(\mathbf{X})$ for this motion.

(b) Compute $\boldsymbol{\omega} = \operatorname{skw}\mathbf{H}$ and $\|\mathbf{w}\|$ (the norm of the axial vector $\mathbf{w} = \operatorname{axl}(\boldsymbol{\omega})$) to determine the rotation field magnitude.

(c) If the small-strain expression is only valid when $|u_{i,j}| < 10^{-4}$, what restriction on the distance to the dislocation core ensures validity? Express your answer in multiples of b.

For this problem, the displacement field $[\boldsymbol{u}] = (0 \quad 0 \quad u_3)$ is provided. The first step towards obtaining the small strain field is to compute the displacement gradient $\boldsymbol{H}$:

$$[\boldsymbol{H}] = \frac{\partial u_i}{\partial X_k} = \frac{b}{2\pi}\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -\frac{1}{1+(X_2/X_1)^2}\frac{1}{X_1^2} & \frac{1}{1+(X_2/X_1)^2}\frac{1}{X_1} & 0 \end{bmatrix}$$

(a) Small strain tensor The small strain can be calculated as per **Eq. (6)** [1]:

$$[\boldsymbol{\epsilon}] = [\frac{1}{2}(\boldsymbol{H}+\boldsymbol{H}^T)] = \frac{b}{4\pi}\begin{bmatrix} 0 & 0 & -\frac{1}{1+(X_2/X_1)^2}\frac{1}{X_1^2} \\ 0 & 0 & \frac{1}{1+(X_2/X_1)^2}\frac{1}{X_1} \\ -\frac{1}{1+(X_2/X_1)^2}\frac{1}{X_1^2} & \frac{1}{1+(X_2/X_1)^2}\frac{1}{X_1} & 0 \end{bmatrix} \tag{6}$$

(b) Infinitesimal rotation tensor $\boldsymbol{\omega}$ The small strain can be compute as per **Eq. (7)**[1]:

$$[\boldsymbol{\omega}] = [\frac{1}{2}(\boldsymbol{H}-\boldsymbol{H}^T)] = \frac{b}{4\pi}\begin{bmatrix} 0 & 0 & \frac{1}{1+(X_2/X_1)^2}\frac{1}{X^2} \\ 0 & 0 & -\frac{1}{1+(X_2/X_1)^2}\frac{1}{X_1} \\ -\frac{1}{1+(X_2/X_1)^2}\frac{1}{X^2} & \frac{1}{1+(X_2/X_1)^2}\frac{1}{X_1} & 0 \end{bmatrix} \tag{7}$$

The vector $\mathbf{w}$ is defined as:

$$\mathbf{w} = \frac{1}{2}\operatorname{curl}\boldsymbol{u} = \frac{1}{2}e_{ijk}\frac{\partial u_k}{\partial X_j} \tag{8}$$

From **Eq. (8)** $u_k = 0$ for $k \,\varepsilon\, [1, 2]$. Thus:

$$\mathbf{w} = \frac{1}{2}\begin{bmatrix} e_{123}\frac{\partial u_3}{\partial X_2} \\ e_{213}\frac{\partial u_3}{\partial X_1} \\ 0 \end{bmatrix} = \frac{b}{4\pi}\begin{bmatrix} \frac{1}{1+(X_2/X_1)^2}\frac{1}{X_1} \\ \frac{1}{1+(X_2/X_1)^2}\frac{1}{X_1^2} \\ 0 \end{bmatrix}$$

The norm of this vector is given by:

$$\|\mathbf{w}\| = \frac{b}{4\pi}\sqrt{\left(\frac{1}{1+(X_2/X_1)^2}\frac{1}{X_1}\right)^2 + \left(\frac{1}{1+(X_2/X_1)^2}\frac{1}{X_1^2}\right)^2}$$

(c) Small Strain Hypothesis Verification The small-strain approximation is valid only if the components of the displacement gradients satisfy:

$$|u_{i,j}| < 10^{-4}.$$

We define the radial distance to the dislocation core as:

$$r = \sqrt{X_1^2 + X_2^2}.$$

From the expression of the displacement gradient tensor $\mathbf{H}$, the nonzero components are:

$$H_{31} = -\frac{b}{2\pi}\cdot\frac{1}{1+(X_2/X_1)^2}\cdot\frac{1}{X_1^2}, \qquad H_{32} = \frac{b}{2\pi}\cdot\frac{1}{1+(X_2/X_1)^2}\cdot\frac{1}{X_1}.$$

These can be rewritten in terms of r as follows:

$$\frac{1}{1+(X_2/X_1)^2} = \frac{X_1^2}{r^2},$$

thus:

$$H_{31} = -\frac{b}{2\pi}\cdot\frac{X_1^2}{r^2}\cdot\frac{1}{X_1^2} = -\frac{b}{2\pi r^2}, \qquad H_{32} = \frac{b}{2\pi}\cdot\frac{X_1^2}{r^2}\cdot\frac{1}{X_1} = \frac{b}{2\pi}\cdot\frac{X_1}{r^2}.$$

To determine which term imposes the most restrictive condition, we analyze the decay behavior when $X_2 = 0 \to X_1 = r$:

- $|H_{31}| \sim \frac{1}{r^2}$ - $|H_{32}| \sim \frac{1}{r}$

Therefore, H_{32} dominates near the core ($r \to 0$). Hence, the ***smallest allowable distance*** r is determined by the condition:

$$|H_{32}| = \left|\frac{b}{2\pi}\cdot\frac{1}{r}\right| < 10^{-4} \quad \Rightarrow \quad r < \left(\frac{1}{2\pi\cdot 10^{-4}}\right) b$$

5 Assignment 5

Assignment 5

Consider a thin-walled cylindrical balloon of length L, diameter D, and wall-thickness T. The balloon is inflated to a length l, diameter d. Assume the deformation is volume-preserving (an incompressible motion) and can be modeled as small strain.

(a) What are the components of the small strain tensor in a *polar* coordinate frame $\{\boldsymbol{e}_r, \boldsymbol{e}_\theta, \boldsymbol{e}_z\}$ which is centered with respect to the undeformed balloon?

(b) What is the thickness of the balloon in the inflated state?

5.1 Small Strain in Cartesian Coordinates

The specified motion can be described in the orthonormal based depicted in **Fig. 3** as:

$$x = \begin{bmatrix} X_1 \frac{d}{D} \\ X_2 \frac{d}{D} \\ X_3 \frac{l}{L} \end{bmatrix}$$

Thus, the displacement and the displacement gradient:

$$[\boldsymbol{u}] = [\boldsymbol{x} - \mathbf{1}] = \begin{bmatrix} X_1(\frac{d}{D} - 1) \\ X_2(\frac{d}{D} - 1) \\ X_3(\frac{l}{L} - 1) \end{bmatrix} \Rightarrow [\boldsymbol{H}] = \begin{bmatrix} \frac{d}{D} - 1 & 0 & 0 \\ 0 & \frac{d}{D} - 1 & 0 \\ 0 & 0 & \frac{l}{L} - 1 \end{bmatrix}$$

Since $\boldsymbol{H}$ is symmetric, the small strain tensor $\boldsymbol{\epsilon}$ is:

$$[\boldsymbol{\epsilon}] = [\boldsymbol{H}] = \begin{bmatrix} \frac{d}{D} - 1 & 0 & 0 \\ 0 & \frac{d}{D} - 1 & 0 \\ 0 & 0 & \frac{l}{L} - 1 \end{bmatrix}$$

5.2 Invariance of the Strain Tensor under Rotation to a Polar Basis

The small strain tensor is initially defined in the orthonormal Cartesian basis $\{\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3\}$ as:

$$[\boldsymbol{\epsilon}]_{\boldsymbol{e}_1\boldsymbol{e}_2\boldsymbol{e}_3} = \begin{bmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & c \end{bmatrix}, \quad \text{with} \quad a = \frac{d}{D} - 1, \quad c = \frac{l}{L} - 1.$$

To express this tensor in a polar coordinate frame aligned with $\{\boldsymbol{e}_r, \boldsymbol{e}_\theta, \boldsymbol{e}_3\}$, a change of basis must be applied. The polar basis vectors are defined in terms of the Cartesian basis as:

$$\boldsymbol{e}_r = \cos\theta\, \boldsymbol{e}_1 + \sin\theta\, \boldsymbol{e}_2, \qquad \boldsymbol{e}_\theta = -\sin\theta\, \boldsymbol{e}_1 + \cos\theta\, \boldsymbol{e}_2, \qquad \boldsymbol{e}_3 = \boldsymbol{e}_3.$$

The associated rotation matrix $\mathbf{Q}$ that maps components from the Cartesian to the polar basis is given by:

$$\mathbf{Q} = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

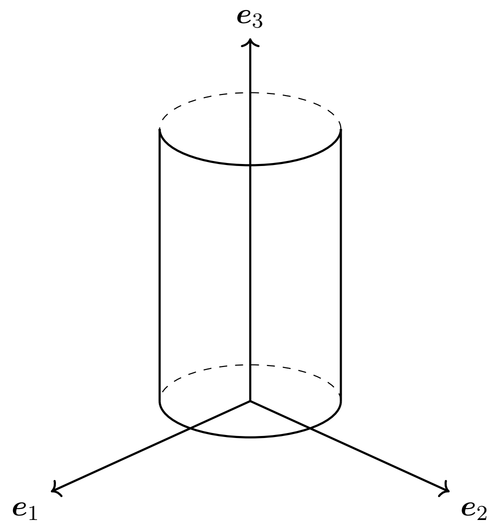

Figure 3: 3D schematic of a thin-walled cylindrical balloon aligned with the cartesian basis.$\{\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3\}$.

The transformation of the strain tensor under this rotation is performed using the standard change-of-basis formula:

$$[\boldsymbol{\epsilon}]_{\boldsymbol{e}_r \boldsymbol{e}_\theta \boldsymbol{e}_3} = \mathbf{Q}\,[\boldsymbol{\epsilon}]_{\boldsymbol{e}_1 \boldsymbol{e}_2 \boldsymbol{e}_3}\,\mathbf{Q}^\top.$$

Substituting the expressions for $[\boldsymbol{\epsilon}]$ and $\mathbf{Q}$ yields:

$$\mathbf{Q}\begin{bmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & c \end{bmatrix}\mathbf{Q}^\top = \begin{bmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & c \end{bmatrix}.$$

It follows that the strain tensor remains diagonal and unchanged when expressed in the polar basis:

$$[\boldsymbol{\epsilon}]_{\boldsymbol{e}_r \boldsymbol{e}_\theta \boldsymbol{e}_3} = \begin{bmatrix} \dfrac{d}{D} - 1 & 0 & 0 \\ 0 & \dfrac{d}{D} - 1 & 0 \\ 0 & 0 & \dfrac{l}{L} - 1 \end{bmatrix}.$$

This invariance is a consequence of the fact that the deformation is axisymmetric and the in-plane components of the strain tensor are equal ($\epsilon_{11} = \epsilon_{22}$). Therefore, the tensor is invariant under any in-plane rotation, including the transformation from Cartesian to polar coordinates.

In order to compute the change in balloon thickness in the inflated state t, the strain in the $\boldsymbol{e_1}$ direction (radial) is computed:

$$\epsilon_{11} = \boldsymbol{e_1}\,\boldsymbol{\epsilon}\cdot\boldsymbol{e_1} = n = \frac{d}{D} - 1$$

The strain in the same direction:

$$\lambda_{11} = \frac{t}{T} = \varepsilon_{11} + 1 = \frac{d}{D} \rightarrow \quad t = T\frac{d}{D}$$

6 Supporting Python Script

A supporting Python script was used to verify manual results and avoid performing cumbersome calculations by hand. The latest version can be found in the GitHub repository.

Python 3.13 Code

```python
import numpy as np

division_lines_count = 30


def run_assignment_2():
    # Parameters
    a0 = 6.0   #
    a = 4.55   #
    b = 5.45   #
    c = 6.02   #
    beta = 87 * np.pi / 180  # radians

    # Cubic lattice vectors
    a1 = a0 * np.array([1, 0, 0])
    a2 = a0 * np.array([0, 1, 0])
    a3 = a0 * np.array([0, 0, 1])

    # Monoclinic lattice vectors
    m1 = a * np.array([
        np.cos(np.pi / 2 - beta) - np.sin(np.pi / 2 - beta),
        np.cos(np.pi / 2 - beta) + np.sin(np.pi / 2 - beta),
        0
    ])
    m2 = c * np.array([0, 1, 0])
    m3 = b * np.array([0, 0, 1])

    # Form matrices with vectors as columns
    A = np.column_stack((a1, a2, a3))
    M = np.column_stack((m1, m2, m3))

    # Obtain F from solving the equation AF = M

    F = M @ np.linalg.inv(A)
    print(f"F= {F}")
    # Compute C = F^T F
    C = F.T @ F
    print("\nC =\n", C)

    eigenvalues, eigenvectors = np.linalg.eig(C)
    print("\nEigenvalues =", eigenvalues)
    print("\nEigenvectors =\n", eigenvectors)

    # Spectral decomposition: C =      _i   r_i      r_i
    U = np.zeros_like(C)

    for i in range(len(eigenvalues)):
        r = eigenvectors[:, i]
        U += np.sqrt(eigenvalues[i]) * np.outer(r, r)
```

```python
    print("\nU (reconstructed from spectral decomposition of C) =\n", U)
    C_verification = U @ U

    # Check closeness
    if np.allclose(C, C_verification):
        print("\nVerification successful: U @ U    C    ")
    else:
        print("\nVerification failed: U @ U does not match C    ")

    E_biot = U - np.identity(3)
    print("\nFINAL RESPONSE\nE_biot =U-1\n", E_biot, f"\nPrincipal strains:\n{[round(float(x)-1,2) for x in eigenvalues]}")


def run_assignment_3():
    print("Assignment 3 Results")

    # Strain tensor
    eps = np.array([
        [1.0, 2.0, 0.0],
        [2.0, 0.5, 0.0],
        [0.0, 0.0, -1.5]
    ])
    print("   =\n", eps)

    # (a) Volumetric strain = trace(  )
    vol_strain = np.trace(eps)
    print("\n(a) Volumetric strain =", vol_strain)

    # (b) Principal strains
    eigvals, eigvecs = np.linalg.eig(eps)
    idx_max, idx_min = np.argmax(eigvals), np.argmin(eigvals)
    print("\n(b) Max principal strain =", eigvals[idx_max], "dir =", eigvecs[:, idx_max])
    print("    Min principal strain =", eigvals[idx_min], "dir =", eigvecs[:, idx_min])

    # (c) Normal strain in n = (1,1,1)/ 3
    n = np.array([1.0, 1.0, 1.0]) / np.sqrt(3)
    nEn = n.T @ eps @ n
    print("\n(c) Normal strain (n) =", nEn)

    # (d) Tensorial shear strain e_nm = 2  n      m
    m = np.array([0.0, 1.0, -1.0]) / np.sqrt(2)
    gamma_nm = (n.T @ eps @ m)
    gamma_mn = (m.T @ eps @ n)
    print("\n(d) Shear strain e_nm =", gamma_nm)
    print("Verifies e_mn=e_nm", np.allclose(gamma_nm, gamma_mn))


if __name__ == "__main__":
    run_assignment_2()
    run_assignment_3()
```

7 References

[1] L. Anand and S. Govindjee, *Continuum Mechanics of Solids.* Oxford University Press, 07 2020.

[2] L. Anand and S. Govindjee, *Example Problems for Continuum Mechanics of Solids.* Independently published, 2020. Publication date: July 25, 2020.

[3] C. R. Harris, K. J. Millman, S. J. van der Walt, R. Gommers, P. Virtanen, D. Cournapeau, E. Wieser, J. Taylor, S. Berg, N. J. Smith, R. Kern, M. Picus, S. Hoyer, M. H. van Kerkwijk, M. Brett, A. Haldane, J. F. del Río, M. Wiebe, P. Peterson, P. Gérard-Marchant, K. Sheppard, T. Reddy, W. Weckesser, H. Abbasi, C. Gohlke, and T. E. Oliphant, "Array programming with NumPy," *Nature*, vol. 585, pp. 357–362, Sept. 2020.